

Supervised Learning, COMP0078
Sample Examination Paper 2 (Level 2)
Suitable for Cohorts: 25/26

Answer ALL EIGHT questions.

Notation: Let $[m] := \{1, \dots, m\}$. We also overload notation so that

$$[\text{pred}] := \begin{cases} 1 & \text{pred is true} \\ 0 & \text{pred is false} \end{cases}$$

Marks for each part of each question are indicated in square brackets.
Calculators are NOT permitted.

1.

- (a) “Ridge regression with a larger regularization parameter λ will tend to lead to underfitting.”
True or False? Explain your answer. [5 marks]
- (b) Consider a binary classification problem with $\mathcal{Y} = \{0, 1\}$. State the Bayes classifier f^* and give an example distribution where the Bayes error is exactly 0.25. Justify your answer. [5 marks]

2.

- (a) We consider properties of kernels.
- (i) Is the following statement true or false? “If k_1 and k_2 are kernels, then $k_1 \cdot k_2$ (pointwise product) is a kernel.” Provide an argument supporting your answer.
- (ii) Is the following statement true or false? “If k is a kernel and $c > 0$ is a constant, then $k + c$ is a kernel.” Provide an argument supporting your answer. [5 marks]
- (b) Let $\mathbf{x}, \mathbf{y} \in \{0, 1\}^n$. Define $K(\mathbf{x}, \mathbf{y}) := \|\mathbf{x} \wedge \mathbf{y}\|_1$ where $(\mathbf{x} \wedge \mathbf{y})_i = \min(x_i, y_i)$ (i.e., the number of positions where both vectors have a 1). Prove that K is a kernel. [10 marks]

3.

- (a) Consider the soft-margin SVM optimization:

$$\min_{\mathbf{w}, \xi} \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^m \xi_i^2$$

subject to $y_i \langle \mathbf{w}, \mathbf{x}_i \rangle \geq 1 - \xi_i, i = 1, \dots, m$.

Compare this formulation to the standard soft-margin SVM. What loss function does this correspond to? Is this formulation still a quadratic program? [5 marks]

- (b) In the hard-margin SVM dual problem, the solution has the form $\mathbf{w}^* = \sum_{i=1}^m \alpha_i y_i \mathbf{x}_i$. Explain what support vectors are and why the optimal hyperplane depends only on them. What does this imply about the sparsity of the solution? [5 marks]

4. In AdaBoost, let D_t denote the distribution over training examples at round t , and let h_t be the weak learner with weighted error ϵ_t .

- (a) Write down how AdaBoost:

- (i) Computes the coefficient α_t for weak learner h_t .
- (ii) Updates the distribution from D_t to D_{t+1} .

Give an intuitive explanation for why the update rule focuses on hard examples. [5 marks]

- (b) Explain the key difference between bagging and boosting in terms of:

- (i) How training data is used for each base learner.
- (ii) Whether base learners are trained independently or sequentially.

Which method is more likely to reduce variance? Which is more likely to reduce bias? [5 marks]

5.

- (a) In the Halving algorithm for learning with expert advice, assume there exists at least one consistent expert among n experts. State the mistake bound of the Halving algorithm and briefly explain the key idea behind its proof. [5 marks]

- (b) The Perceptron algorithm for learning linear classifiers has mistake bound $M \leq R^2 \|\mathbf{u}\|^2 / \gamma^2$ where $R = \max_t \|\mathbf{x}_t\|$, $\mathbf{u}$ is a separator, and γ is the margin.

Consider learning a monotone disjunction of k literals over $\{0, 1\}^n$. What is the mistake bound of Perceptron for this problem? How does it compare to the Winnow algorithm's bound of $O(k \log n)$? [5 marks]

6. Let $G = (V, E)$ be a weighted undirected graph with Laplacian $L = D - W$.

(a) Given the data points $(0, 0), (1, 0), (2, 0), (1, 1)$ in $\mathbb{R}^2$, draw the 1-nearest neighbor graph with respect to Euclidean distance. [5 marks]

(b) Consider a path graph with 5 vertices: $v_1 - v_2 - v_3 - v_4 - v_5$ where all edge weights are 1. Vertex v_1 is labeled +1 and vertex v_5 is labeled -1.

Using the harmonic property (that each unlabeled vertex takes the average of its neighbors' values), find the labels assigned to v_2, v_3, v_4 by Laplacian interpolation. [5 marks]

7.

(a) Define what it means for a hypothesis class H to shatter a set of points. Then define the VC dimension of H . [4 marks]

(b) Consider the hypothesis class of intervals on $\mathbb{R}$:

$$H = \{h_{a,b} : h_{a,b}(x) = \mathbf{1}[a \leq x \leq b], a \leq b \in \mathbb{R}\}$$

Compute the VC dimension of H . Justify your answer by showing both that H can shatter a set of the appropriate size and cannot shatter any larger set. [6 marks]

8.

(a) In the partial feedback (bandit) setting, the learner only observes the loss of the action it selected. Explain how EXP3 constructs an unbiased estimator of the loss vector and why this is necessary. [5 marks]

(b) The Weighted Average (WA) algorithm with learning rate η achieves regret bound:

$$L_{WA}(S) - \min_{i \in [n]} L_i(S) \leq \frac{\ln n}{\eta}$$

for entropic loss.

If we want regret at most $\sqrt{T \ln n}$ after T rounds, how should we set η ? What happens to the regret bound if we don't know T in advance? [5 marks]

END OF PAPER